TM Forum Guidebook

TM Forum Core Frameworks Concepts and Principles

GB991

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Executive Summary

This Core Framework Concepts and Principles document sets the guidelines and best practices that help an enterprise design, plan, implement, and govern the organization of its core building blocks. The structure of the document consists of four main sections: Concepts, Principles, Frameworks (Models) and Methods.

Key concepts defined and described here provide the general ideas formalizing the Open Digital Architecture (ODA). The principles established can help manage an organization's enterprise architecture, with architectural assets such as the Business Process Framework (eTOM), Information Framework (SID), Functional Framework (FF) and Metrics etc. These assets are pivotal to managing and operationalizing the enterprise. Methods breakdown of the complex running of an enterprise into best practices, while the Models provide relational representation of the core business blocks. Altogether, the Framework provides the supporting "beams" to enable an organization to function according to an enterprise architecture blueprint,

This document can be used as a business analysis tool with variations applied to different businesses (e.g. Service Provider). It can also be helpful when clarifying concepts to organize ideas. By leveraging a solid framework, as described in this document, the reader is able to capture "realistically" core business concepts of an organization and do this in a way that is tried and proven.

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Introduction

The key concepts described in this document provide a common language and understanding for the architect, other associates and business stakeholders. The foundational CONTEXTS are used to establish the enterprise's building blocks. These building blocks are fundamental to the Open Digital Framework (ODF) as they define key concepts that are used throughout ODF. The core frameworks that make up the TM Forum ODF include the Business Process Framework (aka eTOM), the Information Framework (aka as SID), and the Functional Framework (derived from the previous TAM).

The key principles provide the rules and guidelines that govern the application of key concepts in the Open Digital Framework. The rules are designed to reflect the values and goals of a digital organization, while providing direction and guidance for decision-making. These principles help to ensure consistency, alignment, and coherence across any architecture that aspires to align with the Open Digital Architecture (ODA).

The section on method or methodology establish the processes and techniques that help an enterprise architect leverage the Open Digital Framework. They provide a structured and systematic approach to identify, analyze, design, and communicate the architecture. These methods or methodologies are here to help ensure quality, efficiency, and effectiveness of the target architecture of the organization.

Models are the representations and artifacts that help the enterprise architect, design, document and communicate the architecture. In TM Forum, the Master Open Digital Architecture (MODA) is the definitive reference for ODA which provides the context and visualization of the structure, behavior, and relationships of the ODA business objects (architectural elements). Models help to ensure clarity, purpose, completeness and consistency of the architecture as they evolve through a maturation process which improves its reliability for TM Forum members.

The adoption of the TM Forum core frameworks has the benefit of providing alignment with key industry standards and common ontology, in support of the enterprise's strategic and business goals. This can help boost both business and technical capabilities. All principles described in this document provide a foundational ground for the development of the enterprise's governance, rules and policies.

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1. Concepts and Definitions

1.1. Contexts

Contexts provide an overarching structure for the frameworks for the classification and grouping of all business objects and architectural artifacts. There are two types of contexts; 1) Horizontal Contexts commonly known as "Domains", and 2) Vertical Contexts.

Not all domains and vertical contexts are present in each of the frameworks

1.1.1. Horizontal Context (Domains)

Horizontal contexts are commonly known as Domains are of two kinds 1. Business Domains (shown in color in the diagram below) and 2. Non-Business Domains.

The Business Domains represent areas of operations for delivery and management of the business.

The Non-Business Domains represent collections of common objects and technical functions supporting the business areas.

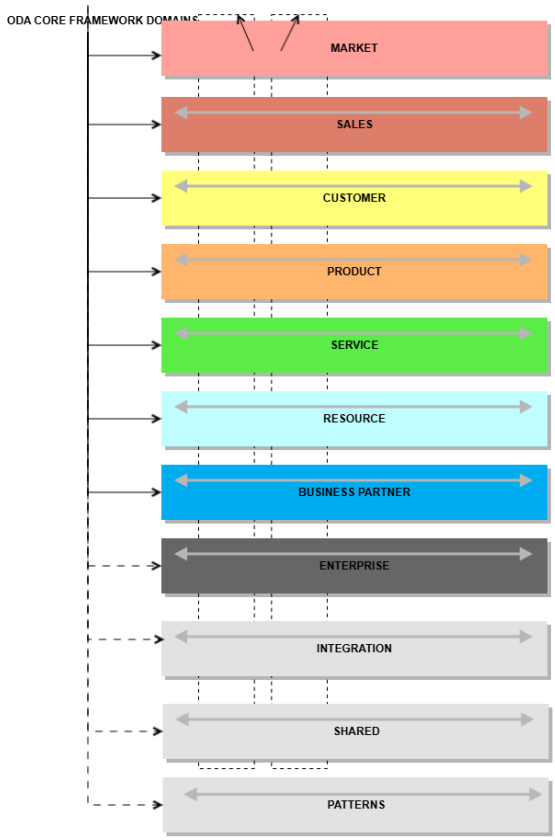

Figure 1-1 Core Frameworks Domains

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1.1.1.1. Market Domain

The Market domain focuses on understanding, shaping, and influencing demand by managing market intelligence, brand strategy, customer insight, and go-to-market planning. It encompasses activities that position the organization in the market, develop marketing strategies, conduct research and segmentation, manage brand and reputation, and design outbound campaigns to drive awareness and interest. The Market domain translates the organizations' goals into market-facing strategies. The domain also includes the continuous assessment of market performance, emerging trends, and customer behavior to guide strategic decisions and shape the development of products and offerings.

1.1.1.2. Sles Domain

The Sales domain focuses on the execution of sales strategies to generate revenue by engaging with prospects and customers, managing opportunities, and closing deals. It includes activities that convert marketing interest into actionable leads, qualify and manage sales opportunities, negotiate commercial terms, and drive deal closure across channels. The domain ensures that customer needs are matched with suitable product offerings and supports ongoing sales development and performance tracking. Sales acts as the interface between customer demand and commercial delivery, operating through direct partner, and digital sales channels.

1.1.1.3. Product Domain

The Product domain represents roles, activities, information and functions carried out by parties (e.g., individuals / organizations and things) playing roles that are involved in the strategic planning, definition, development and operational aspects of Products that are conceived and offered for marketing and sales by the Service Provider.

1.1.1.4. Customer Domain

The Customer domain represents roles, activities, information and functions carried out by parties (e.g., individuals / organizations) playing roles that are involved in the management of and all types of contact with customers as they acquire, use, pay for and are supported for goods and services (i.e., products) that they obtain from an enterprise. Activities include: Strategy to Readiness (e.g., customer strategies, capabilities, customer lifecycle management) and Operations (e.g., customer relationship management , data, privacy, interactions, communications, orders, accounts, balances, service level agreements (SLAs), training, problems, cases, invoices, payments, disputes, collections, loyalty, performance, usage statistics, analytics and support).

1.1.1.5. Service Domain

The Service domain represents roles, activities, information and functions carried out by parties (e.g., individuals / organizations) playing roles that are involved in the strategic planning, definition, development, and operational aspects of Services that are used to realize Product offerings to the market. Activities include management of: strategies, capabilities, lifecycles, catalogs, inventories, installations, activations, problems, performance, guiding, mediation, usage statistics and support of customer-facing services that are offered to customers and resource-facing services that are presented to resources by an enterprise.

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1.1.1.6. Resource Domain

The Resource domain represents roles, activities, information and functions carried out by parties (e.g., individuals / organizations) playing roles that are involved in the strategic planning, definition, development and operational aspects of Resources (e.g., functions, applications, computing, networking and storage) that represent the infrastructure of an enterprise that are used to realize Services. Activities include management of strategies, capabilities, lifecycles, catalogs, inventories, topologies, installations, activations, alarms, problems, performance, mediation, usage statistics and support of Resources that are managed by an enterprise.

1.1.1.7. Business Partner Domain

The Business Partner domain represents roles, activities, information and functions carried out by parties (e.g., individuals / organizations) playing roles that are involved in the strategic planning, definition, development, operational aspects and all types of contact with Business Partners (e.g., Suppliers, Partners, etc.) with which an enterprise collaborates in order to operate their business. Activities include management of Business Partner: strategies, capabilities, value propositions, relationships, profiles, data, privacy, security, interactions, communications, tenders, agreements, orders, requisitions, supplies, accounts, balances, inventories, reconciliations, service level agreements (SLAs), training, problems, cases, invoices, payments, revenues, disputes, collections, loyalty, performance, usage statistics, analytics and support of Business Partners as they supply, acquire, use, support, purchase, pay for and are supported for goods and services (products) that they provide and / or obtain from an enterprise.

1.1.1.8. Enterprise Domain

The Enterprise domain represents roles, activities, information and functions that are required to run and support a business. These concepts focus on both the setting and achieving of strategic corporate goals and objectives, as well as providing those support services that are required throughout an Enterprise. These concepts are sometimes considered to be the corporate functions and/or processes (e.g., Financial Management, Human Resources Management processes, etc.). Since Enterprise Management is aimed at general support within the Enterprise, they may interface as needed with almost every other process in the Enterprise, be they operational, strategy, infrastructure or product processes.

1.1.1.9. Shared Domain

The Shared domain is a specialized domain that is formalized for use in the Information Framework (SID) and the Functional Framework. This special domain represents information, functions and metrics that are not “owned by” any particular domain and are referenced or utilized from two or more other domains.

1.1.1.10. Patterns Domain

The Pattern domain is a specialized domain that is formalized for use in the Information Framework. It contains abstract ABEs not directly usable. These patterns need to be sub-classed to be used.

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1.1.1.11. Integration Domain

The integration domain represents business-agnostic roles, information and activities (e.g., infrastructure, middleware, protocols) that support the integration and implementation of ODF concepts (e.g., business entities, business processes, functions, logical applications, interfaces, metrics, etc.).

1.1.2. Vertical Context (Lifecycle Stage)

Vertical contexts are groupings of elements of the respective frameworks for Development, Implementation and Operation of business entities and business artifacts related to the respective horizontal contexts.

1.1.2.1. Vertical Context Area (Lifecycle Area)

Vertical Context Area provide a classification of vertical contexts mostly relevant to the Business Process Framework (eTOM) and the Functional Framework; they represent a separation of concerns by dividing the enterprise lifecycle in two parts; 1) Strategy-to-Readiness (includes 'Strategy Management', 'Capability Management', 'Business Value Development' and a shared part of 'Operations Readiness & Support'), and 2) Operations (includes a shared part of 'Operations Readiness & Support', 'Fulfillment', 'Assurance', and 'Billing').

1.1.2.2. Vertical Context Stage (Lifecycle Stage)

Vertical Context Stage provide a classification of vertical contexts as a means to further group them according to the two parts of the enterprise lifecycle, as described previously.

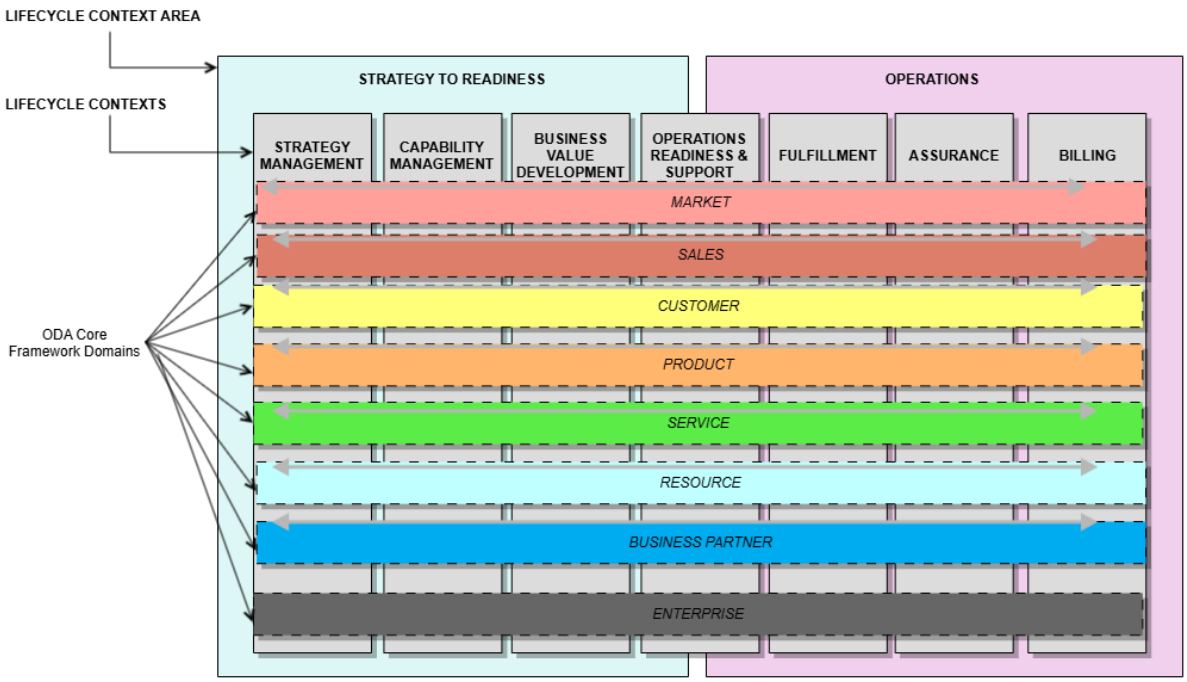

Figure 1-2 Core Frameworks Contexts with Domains overlaid

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Strategy-to-Readiness Lifecycle context area

Reference to section 1.1.2.1, the Strategy-to-Readiness Lifecycle context area represent the grouping of the lifecycle Context Stages: Strategy Management, Capability Management, Business Value Development, and part of Operations Readiness & Support. (see Figure 1.1).

Strategy-to-Readiness" area identifies the business processes that can help organizations prepare for engagement with their markets. It includes transformation that readies people, processes and technology/platforms for operations.

Strategy Management

The Strategy Management vertical is a lifecycle stage that focuses on business objects that generate and manage specific business strategies, gain business buy-in, track effectiveness of the strategies, and optimize strategy required.

Strategy Management is focused on the analysis and generation of strategies in support of all development and operations within the enterprise while establishing business commitment within the enterprise.

Capability Management

The Capability Management vertical is a lifecycle stage that focuses on the development and deployment of capabilities including all infrastructures for the delivery of products and services as well as support infrastructures and all business enablers.

Framework artifacts in this vertical grouping identify requirements, design and develop new or enhanced capabilities and monitor their performance to adjust as required.

Business Value Development

The Business Value Development vertical grouping focuses on the definition, planning, design and implementation of value propositions and business promoting concepts/assets in the enterprise's business portfolio.

Frameworks artifacts in this vertical grouping create products, services, resources, business partner catalog information, sales and marketing campaigns and material and process Operations feedback to adjusts as required for Lifecycle management.

Operations Lifecycle context area

The Operations context area groups the lifecycle context stages: Fulfillment, Assurance, Billing, and part of Operations Readiness & Support.

The Operations context area includes all the artifacts and building blocks that support customer operations, as well those that enable direct interactions with the customer. Such artifacts include both day-to-day and operations readiness and support. The "FAB" (or "OFAB") verticals in Operations are sometimes referred to as Customer Operations processes.

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Operations Readiness & Support

Operations Readiness & Support lifecycle stage is responsible for ensuring operational readiness and support of Fulfillment, Assurance and Billing.

Fulfillment

Fulfillment lifecycle stage is responsible for initiation, instantiation and fulfillment of customer requests.

Frameworks artifacts in this vertical grouping initiate fulfillment actions and convert requests (customer or business) into orders, to be delivered using the value offerings from the enterprise's portfolio. This includes creation of instances of relevant artifacts in all domains necessary for the delivery of the value offerings.

Assurance

This vertical context is responsible for running operations along with monitoring, measuring, analyzing and correcting to ensure the best performance, including Quality of Service. Frameworks artifacts in this vertical context perform continuous status and performance monitoring of ongoing operations. They collect performance data and analyze them to identify potential problems and apply preventive and corrective maintenance without impacting the customer.

The Assurance activities receive trouble reports and take necessary actions to ensure timely corrections.

The Assurance activities ensure that overall performance and quality meets service level agreements, metric goals (KPIs) as well as reporting.

Billing

This vertical context is responsible for translating business activities into revenue.

Framework artifacts in this vertical context perform collection of appropriate usage records to determine charging and production of timely and accurate bills and supports payment of services and operation-related cost accounting.

They provide pre-bill usage information and are processing customer payments and performing payment collections.

1.2. Business Capability Map

(FUTURE WORK)

1.3. Business Process Framework

The Business Process Framework is a reference framework or model for the definition, classification and decomposition of Service Provider business process abstractions. A business process abstraction is a generalization of actual *context-specific business processes*. So, rather than describing the process as it exists at company A or company B, or in one part of an organization versus another, the abstraction describes the essence of the process, and as such it can be used to

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"categorize" any or all of those context-specific processes that match the description.

For more than 20 years, the TM Forum has developed the Business Process Framework in collaboration with its members, and has achieved industry-wide recognition, from the highest conceptual concepts, to the detailed process decomposition, taking into account real-world experience in the design, implementation and management of business processes in a service provider's environment and operations. It represents an ontology of business activities without prejudice to specific business process models or workflows realized in the context of specific enterprise strategies and market dynamics.

1.3.1. Purpose

The purpose of the Business Process Framework is to provide a standard process structure, terminology, and classification scheme for business processes in an enterprise. It offers an enterprise-wide discipline for the development of business processes that assure a consistent and reusable repository of processes in business process models.

As a framework, it serves as a tool for discussion and agreement between providers and consumers which encourage clear definitions, convergence and understanding of:

- Business requirements
- Information agreements
- Application contracts; and
- shared data model specifications (exchanges between applications or systems)

TM Forum BPF is a cross-industry process view that presents a common ontology for equipment suppliers, applications builders and integrators to define, build and operate systems and solution.

1.3.2. Scope

TM Forum Business Process Framework recognizes the different types of business processes in an enterprise. The scope of the Business Process Framework covers the three broad categorization of an enterprise business processes - Core, Enabling and Supporting (also categorized in other ontologies as Operational, Management and Supporting respectively.)

1.3.2.1. Core Process

Core Process is a business process that constitute the core business and create the primary value stream of an enterprise. They are strategically important to the enterprise, and derive from the core competency of the business the enterprise is involved in -like taking and delivering orders from customers, defining and delivering products and services etc.

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1.3.2.2. Enabling Process

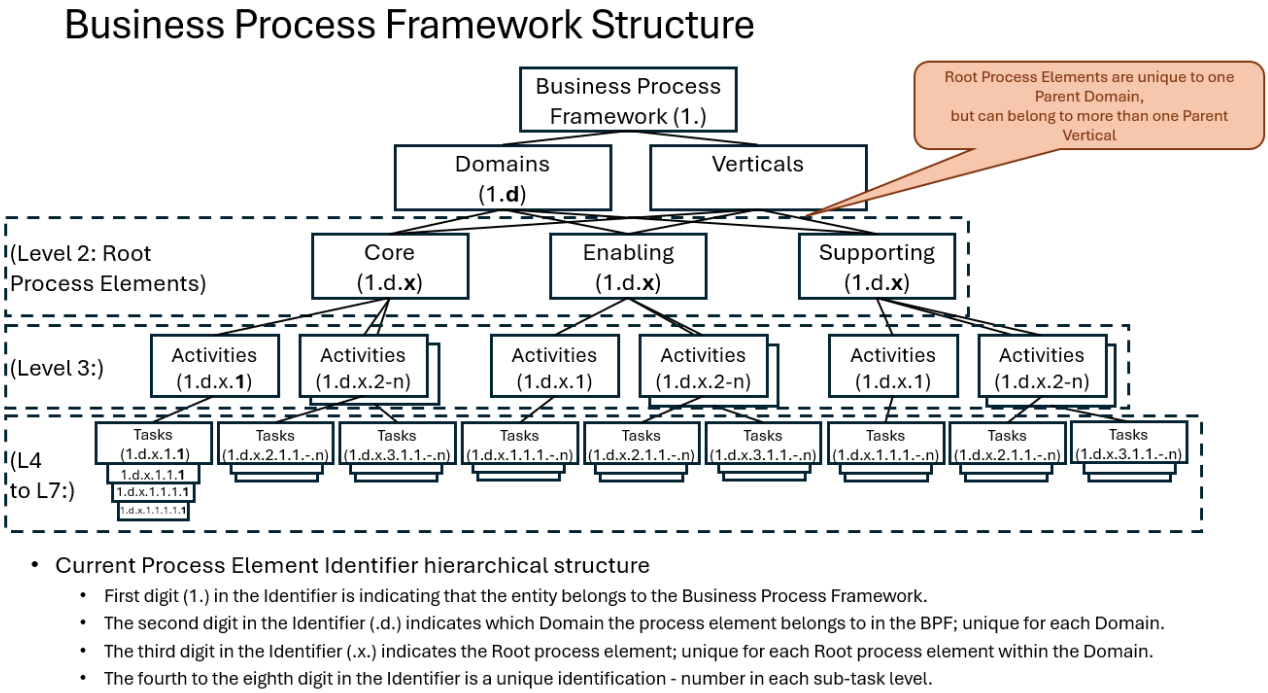
Enabling Process, also known as management processes, is a business process that sets direction, like monitoring performance and executing plans for the enterprise.

1.3.2.3. Supporting Process

Support process is a business process that services the basic functions of the organization, like accounting, technical support, human resource etc.

1.3.3. Conceptual View

The conceptual view below shows the decomposition structure for a business process, from the top level to lower levels. At the highest level of representation is the "Business Process". The decomposition of the "Business Process" in directly supporting "Business Activity" starts at the level 3.



- Current Process Element Identifier hierarchical structure
  - First digit (1.) in the Identifier is indicating that the entity belongs to the Business Process Framework.
  - The second digit in the Identifier (.d.) indicates which Domain the process element belongs to in the BPF; unique for each Domain.
  - The third digit in the Identifier (.x.) indicates the Root process element; unique for each Root process element within the Domain.
  - The fourth to the eighth digit in the Identifier is a unique identification - number in each sub-task level.

Figure 1-3 Conceptional organization of TM Forum Business Process

1.3.4. Other Concepts

1.3.4.1. Business Process Model (Workflow)

A business process model is a graphical representation of a business process or workflow and its related sub-processes. It defines the ways in which operations are carried out to accomplish the intended objectives of an organization. Business process modeling is the graphical representation of a company's business processes or workflows. This is usually done through different graphing methods, such as the flowchart, data-flow diagram, etc.

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Refer to Section 4 in this document for more information about the methodologies used. See specific examples of Business Process Models in GB921F and GB921E where examples showcasing specific uses of Business Process Framework ontology to establish business process models that show E2E workflows depicting for "product management" and "order management" etc.

1.4. Information Framework

The Information Framework is an abstract but formal / defined representation of business entities including their properties and relationships that can be performed on them. A business entity is a thing of interest to the business, while its attributes are facts describe the entity. The formal definitions provide a business-oriented perspective to the information and data with a relationship between information and data representing interests between two business entities, or between a business entity and itself.

The TM Forum Information Framework provides an industry reference for representation of concepts and the relationships, constraints, rules, and operations to specify information semantics for a chosen domain of discourse. It typically specifies relations between kinds of things but may also include relations with individual things.

1.4.1. Purpose

TM Forum Information Framework provides sharable, stable, and organized structure of information requirements or knowledge for domain contexts. As a framework, it serves the purpose of providing an enterprise with information definitions along with underlying relations between information exchanged by business processes.

In combination with the Business Process Framework, the Information Framework offers a way to explain ‘how’ things are intended to fit together to meet a given business need.

1.4.2. Scope

The scope of the TM Forum Information Framework is to:

- Establish an enterprises approach to Information management and governance. This includes the aggregating closely related business entities and governing the relationships between business entities.
- Describe the holistic approach to managing information created/read/updated/deleted by business processes through roles and metrics to transform information into business asset
- Standardize the grouping of business entities into Aggregate Business Entities
- Formalize Relationship between Business Entities

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- Identify and associate a common set of key facts that can be related to business entities

1.4.2.1. Aggregate Business Entity

An Aggregate Business Entity (ABE) is a well-defined set of information and operations that characterize a highly cohesive, loosely coupled set of business entities.

In information modeling, an `ABE` is a way to further partition `DOMAIN` by grouping related `Business Entities` (BEs) together. `ABE` can contain other `ABE` or `BE` and can be considered as sub-domains. A given domain can contain anything from a few to over 15 ABEs.

1.4.2.2. Business Entity

A Business Entity represents something of interest to the business that may be tangible things (such as a customer), active things (such as a Customer Order), or conceptual things (such as a Customer Account). Business entities are characterized by attributes and participate in relationships with other business entities. Business entity instances typically move through a well-defined life cycle.

1.4.2.3. Attribute

An attribute is a fact that describes a business entity.

1.4.2.4. Relationship

A relationship is an association of business interest between two business entities, or between a business entity and itself.

1.4.3. Conceptual view

At the highest level, the Information Framework is structured according to the ODA horizontal contexts (aka domains). Within each domain there is a high degree of cohesion. Within each domain there is a high degree of cohesion between the identified business entities, and loose coupling between different domains. This enables segmentation of the total business problem and allows resources to be focused on a particular domain of interest. It is envisioned that the use of the resultant business entity definitions within each domain, when used in conjunction with the Business Process Framework, will provide a business view of the shared information. Within each domain, further partitioning of the information is achieved through the identification of Aggregate Business Entities (ABE's).

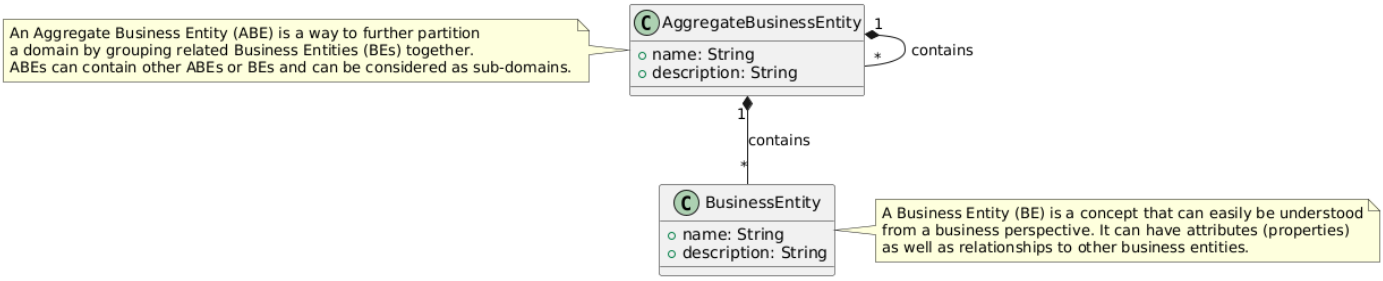

Figure 1-4 Conceptual organization of Information Framework Framework

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1.5. Functional Framework

The Functional Framework is a collection (an ontology) of functions as defined in 1.5.2.1 organized in a hierarchical fashion based on functional affinity. This organization is based on domains, verticals, and aggregate functions (AFs).

As the Functional Framework is an ontology, a given function is unique within the Functional Framework.

1.5.1. Intended Uses

Use by members (service providers, software providers, etc) can include:

- Rationalization of systems
- Governance
- A common language for describing systems and system elements,
- Basis for service provider to discuss with a software provider about solutions scope and functional coverage
- Organize and rationalize systems elements, including applications, components, APIs, etc.

TMF Use:

- Bridge between business-level concepts (Processes, Business Capabilities) and systems-level concepts (API, Components)
- Identify and describe the functions that are implemented by each component
- Identify gaps and opportunities for coverage in definition of the Open APIs

1.5.2. Scope

The scope of the functional framework includes anything that the *systems* supporting the Telecom Industry may need to do. The Functional Framework scope is intended to include functionality of Business Support Systems (BSS), Operational Support Systems (OSS), Enterprise Systems, security and network elements within the Telecommunications industry. This includes the non-business technical aspects.

1.5.2.1. Functions

A function is a defined set of action(s) that a system is able to do in support of business or technical needs. A function is defined as cohesive around a focused purpose. Functions are typically characterized by action-oriented features, and often focus on elements of information. Functions are implementation-agnostic (identifying what systems do as opposed to how).

1.5.2.2. Aggregate Functions

An aggregate function is a cohesive grouping of functions. The description of each aggregate function should make clear on what basis functions are grouped together within that aggregate function, and should provide sufficient basis to

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determine whether a given function would belong within or outside of that Aggregate Function.

Aggregate Function Level 1
Aggregate Functions Level 1 contain Aggregate Functions Level 2 or functions.

Aggregate Function Level 2
Aggregate Functions Level 2 contain Functions.

1.5.3. Conceptual view

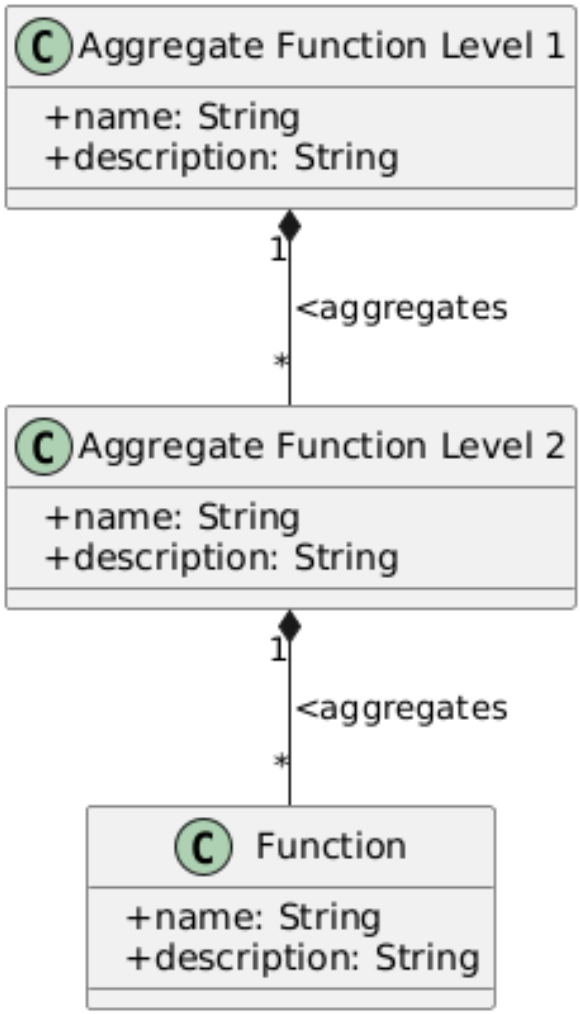

Figure 1-5 Conceptual organization of Functional Framework

1.6. Metrics

(FUTURE WORK)

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2. Principles

2.1. Shared

S-01	***One framework organizes all ODA Core objects***
Statement	A shared framework is a common language and structure that enables different parts of the Open Digital Architecture to organize concepts that work together effectively towards same goals, by improving communication and collaboration, and by reducing confusion and misunderstandings.
Rationale	• Improve communication and collaboration between different parts of the enterprise. • Better decision-making with context based on bounded zones / blocks. • Provide a common language and structure for different foundations of ODA to work together effectively. • Ensure producers and consumers are working towards the same goals with reduced confusion and misunderstandings.
Implications	• Provide a common language and structure for Open Digital Architecture foundation assets that serve as primitives for enterprises to plan, transform and operate. • Reduce confusion and misunderstandings between different departments or teams and help to ensure that business objects grouped are altogether supporting same goals. • Improve communication and collaboration between different parts of an enterprise, which can lead to better decision-making and more effective problem-solving.

S-02	***Release and Publication Status***
Statement	The shared release and publication status for all ODA Core assets shall be one of: • Released • Preliminary • Draft • Not Fully Developed
Rationale	• Applying the same lifecycle statuses across all elements promotes consistency, predictability, and transparency. It allows for better tracking, management, and understanding of the state of each element at any given time.
Implications	• Ensures that all stakeholders have a clear understanding of the status of each element.

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S-02	***Release and Publication Status***
	• Facilitates better planning and resource allocation based on the lifecycle status. • Promotes accountability as each status requires specific actions and responsibilities. • Enhances efficiency by standardizing the process of updates and releases.

S-03	***Business-first understanding***
Statement	All business processes, information models, and software functions shall be designed with simplicity and a business-first approach. They should be easily understandable and directly contribute to the achievement of business goals.
Rationale	Simplicity leads to better understanding, efficient execution, and easier maintenance. A business-first approach ensures that all processes, models, and functions are aligned with the enterprises objectives and contribute to its success.
Implications	Keep models simple and understandable by Business first, and then technology personas. This will: • Facilitate clear communication across all levels of the enterprise. • Enhance efficiency by reducing complexity in processes, models, and functions. • Ensure alignment of all operations with the enterprises' business goals. • Promotes a culture of simplicity and clarity in the enterprise.

2.2. Framework Specific

2.2.1. Business Capability

(FUTURE WORK)

2.2.2. Business Process Framework

BPF-01	**Nature of the Business Process Framework**
Statement	The Business Process Framework consists of a decomposition structure and taxonomy of process abstractions representing business process elements (i.e. groups of processes, activities and tasks) that a Service Provider needs to manage.
Rationale	The Business Process Framework is intended to be comprehensive - all of a service provider's processes, activities and tasks should fit within the Business Process Framework.

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BPF-01	**Nature of the Business Process Framework**
Implications	The process abstractions of the business process framework are categorizations of the context-specific processes that exist within service providers. The Business Process Framework provides a comprehensive catalog of such process abstractions relevant for a Service Provider. It provides an industry agreed standard consisting of process decomposition structure, terminology, and classification scheme of process elements across multiple levels of decomposition. The process elements of the Business Process Framework can be used to create consistent process flows (however, they are not business process flows themselves) The business drivers for adopting and using the framework can be summarized according to the following: • The need to standardize the language for describing business processes • The need to standardize tasks on which business process flows will be based • The need to understand the elements that make up each business process. • The need to understand the objectives associated with each business processes

BPF-02	**Standards for Organizing Business Processes - Naming, Decomposition and Traceability**
Statement	A business process element shall be named and decomposed according to the following pattern or structure. Naming: • Noun verb (for a business process grouping) • Verb noun (for underlying activities and tasks that support the business process) Decomposition: • The Business Process Framework is decomposed from notional Level 0 to more granular levels • The Levels 1, 2, 3 and 4 (and some of levels 5 and 6) are addressed so far.
Rationale	This prevents any confusion and establishes a unified level setting for business processes.

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BPF-02	**Standards for Organizing Business Processes - Naming, Decomposition and Traceability**
	Because the Business Process Framework is a decomposition model, the lower levels of the decomposition can be traced back to the higher levels. As a classification or taxonomy of Process Elements, at Level 0 the elements are classified into S2R (Strategy to Operations) and Operations. Lower Levels are formed by decomposition with each Process Element occurring once only.
Implications	• Achieve consistency in naming and decomposition following a consistent pattern • Ensure the goal is implied in the name of the business process with relevance primary stakeholders. • Ensure the name resonates with industry use or advances industry consensus about the primary role the business process plays.

BPF-03	**Classification of Business Processes**
Statement	All business processes shall be classified as one of the following types: • Core Process • Enabling Process • Supporting Process
Rationale	Enable organizations to identify important business actions that impact their core business, enabling the management of resources and the efficiency of support. This includes cost reduction and resource allocation.
Implications	• Enable enterprises to understand and manage business processes more effectively. • Identify levels of importance of business processes and allocate associated resources accordingly. • Help to improve efficiency and reduce costs of operationalizing business processes, while also ensuring that the enterprise is focused on its core business goals.

BPF-04	**Definition of Business Processes**
Statement	All business processes should be clearly defined in terms of their structure, constituents, roles and purpose.
Rationale	Enable organizations to ensure their business processes are consistent and complete.

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BPF-04	**Definition of Business Processes**
Implications	• A business process has characteristics that include goals, inputs, outputs and business outcome. • A well identified, scoped and articulated business process must provide clarity on the resources underpinning its activities. • Organizations should be able to realize goals linked to business processes by formalizing them in specific sequence to realize value.

BPF-05	**Information Ownership By Business Processes**
Statement	Each Business Process element should be associated with one (or more) Information Framework ABE(s).
Rationale	To be consistent, processes must be related to specific information elements.
Implications	• Business processes do not exist in isolation. Processes require information from other processes, and they in turn provide information to other processes. • Dependencies (or associations) between processes occur when an activity requires information from another activity. Process dependencies are related to the entities and attributes required by the business area.

BPF-06	**Business Process Ownership**
Statement	A business process shall always have a “Process Owner” who acts as the person or instance that has the highest level of responsibility and decision-making with respect to the process; such person or instance is designated as being accountable for the outcome that results from the execution of the process.
Rationale	Responsibility can be shared amongst individuals involved in the execution of a particular process, however, accountability–which designates ultimate responsibility and ownership of the process–refers to the unique instance or person who is in charge of the entire process, and therefore accountable for the outcome resulting from the execution of such process, this instance or person is designated as the “accountable” party and it is in general referred to as “the Owner” of the process.
Implications	• "Shared responsibility is no accountability”; • a business process is always overseen and “groomed” by its owner who is the person or instance ultimately responsible for the outcome of the execution of the process; • failure to designate or have implicitly a process owner or accountable party would result in lack of commitment or

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BPF-06	**Business Process Ownership**
	accountability in the event that the process is not delivering the expected outcome; likewise, if the process is delivering excellent results, it is most often the process owner (or accountable instance) that gets glorified.

BPF-07	**Continuously Optimize Business Processes**
Statement	New information systems will be implemented after business processes have been analyzed, simplified or otherwise redesigned as appropriate.
Rationale	• There is no real "value" in applying technology to old, inefficient business processes. • Core processes will be more streamlined, efficient and cost-effective. • Core processes, activities, tasks and associated business rules will be well understood and documented. • Reduces the total cost of ownership.
Implications	• Need to have an agreed upon continuous business process improvement activity. • New technology will be applied in conjunction with business process review. • Business processes must be optimized to align with business drivers. • Additional time and resources will have to be invested in analysis early in the systems' lifecycle. • Organizational change will be required to implement reengineered work processes. • May require regulatory or legislative change.

BPF-08	**A core process has a unique position in the process framework**
Statement	All Level-2 (L2) business processes have a specific, unique position in the process framework, their position depends on the purpose, nature, and role of the process within the enterprise. As a general rule, each position is determined by the intersection between the vertical grouping and the horizontal domain to which the process belongs in the process framework. In general, L2 process belong to one unique vertical and one unique horizontal domain; however, there are some exceptions in which some L2 processes can span more than one vertical, these are commonly referred to as cross-functional or transversal core processes.
Rationale	The Business Process Framework was created and evolved from two different but complementary models: 1) An Enterprise Architecture layering model (based on a construct of hierarchical architecture layers) which originated with the early concepts underpinning the TMN model; and 2) An enterprise lifecycle structure consisting of vertical groupings

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BPF-08	**A core process has a unique position in the process framework**
	(aka lifecycle stages) which originated in the early concepts associated with the FCAPS model. Such design resulted in a matrix of X and Y positions, each position characterized by a specific couple 'vertical/horizontal', representing a unique stage in the enterprise lifecycle and a unique horizontal architecture domain; as a general rule, each position hosts several L2 processes, therefore, each L2 process belongs to one unique vertical and one unique horizontal domain; however, some L2 processes span more than one vertical, these are referred to as cross-functional or transversal processes, they are considered exceptions to the previous rule, and therefore they are prone to change in the future as the framework continues to evolve.
Implications	Level-2 (L2) processes play a vital role in the lifecycle of the enterprise, each L2 process represents a grouping of activities and tasks that deliver the intended business outcomes for each part of the enterprise process framework, as such, L2 processes are designed based on the role they perform and outcome they deliver, both of which are determined by the position of the process in the framework i.e. vertical grouping(s) and horizontal domain to which the process belongs. In the past, common process models were designed contextless–that is–core processes were monolithic elements decomposed into smaller granularity elements, but without being directly related or mapped to any specific context i.e., didn't belong to any particular enterprise lifecycle or architecture domain e.g., Product, Service, Resource, etc. The Business Process Framework (eTOM) brought along an innovative classification and modelling of the business processes across the enterprise, by adding horizontal domains and vertical Lifecycle stages which provided a meaningful architectural context.

BPF-09	**Business Processes lead on architecture requirements**
Statement	Business processes represent an implementation of the business requirements sourced from the organization's strategy and business intent, as such, business processes embody the required support and full alignment with the business to ensure business operations and their goals are achieved.
Rationale	Business processes embody the activities and tasks that need to be in place, to enable and support successfully the operation of the enterprise, business processes are in alignment with the goals of the business, as such, they translate business intent and required business capabilities into "implied" functional requirements that are further communicated to the functional, application and technology architectures–as subordinates of the Business Architecture–for implementation and execution through either manual or automated support, or a combination of both.
Implications	Business processes play a central role in the business architecture, as they define altogether the fabric of activities and tasks that are needed to effectively support and run the business. It is a vital principle to

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BPF-09	**Business Processes lead on architecture requirements**
	always ensure an optimal alignment between the enterprise business goals, and the capabilities available to support and achieve those goals.

2.2.3. Information Framework (Kevin/Cecile- September 7)

#INF-01	***Coupling of ABEs to Business Processes***
Statement	An ABE shall be created by one business process (denoted primary process). If a one-to-one coupling is not achievable, an ABE shall be coupled to associated
Rationale	A single primary process should manage the complete life cycle of an ABE, creating, updating, and deleting entity instances contained within the ABE.
Implications	If there is more than one process creating an ABE, this suggests these are overlapping processes, or the scope of the ABE is too broad.

#INF-05	**ABEs, Core Entities and Dependent Entities**
Statement	There should be one single core business entity in an ABE in general (there are some exceptions however), in most cases the core entity has the same name as its parent ABE.
Rationale	A well-defined Aggregate Business Entity (ABE) consists of a set of information that characterizes a highly cohesive, loosely coupled set of business entities, which have loose relationships with items outside the ABE.
Implications	A core business entity is an entity that is not dependent upon any other entity within the ABE. Likewise, generally, non-core business entities are dependent on the core business entity. And do not need to have an 'ID' attribute, given they can be reached by navigation from the core business entity.

#INF-06	**Principles for ABE Placement in Shared Domain versus Patterns Domain**
Statement	ABEs and related BEs will be placed in Shared when: 1. One or more "Core" Business Entities are "Concrete", AND 2. Business Entities must be used in more than one domain without modification or subclassing (e.g. - Party) A core entity is a main entity (not a subclass) ABEs and related BEs will be placed in the Patterns Domain when: 1. All "core" business entities are *abstract*. Business entities, attributes, and/or relationships are inherited by concrete business entities within other ABEs.
Rationale	To be completed
Implications	To be completed

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#INF-07	Principles for ABE Placement in Patterns Domain
Statement	ABEs and related BEs will be placed in the Patterns Domain when 1. All "core" business entities are *abstract*. 2. Business entities, attributes, and/or relationships are inherited by concrete business entities within other ABEs.
Rationale	To be completed
Implications	To be completed

#INF-08	Inheritance
Statement	The Information Model is built on the principle of "single inheritance". Not all business entities need to be "rooted", but a given class/business entity can only inherit from one entity. Notes: ▪ Must verify that this is true ▪ One reason for not doing multi-inheritance is because it can be a challenge to 'implement'
Rationale	To be completed
Implications	To be completed

#INF-09	Association Directionality
Statement	Within the SID model, an association is assumed to be bi-directional.
Rationale	To be completed
Implications	To be completed

#INF-10	Ennumerations
Statement	Generally, the Information model does not use enumerations. Attributes with some number of given values is handled as follows:
Rationale	To be completed
Implications	To be completed

2.2.4. Functional Framework

Please note!

We are currently in the process of redefining the principles of the Functional Framework. This is an iterative process and we plan

to remove the current principles and add new one as they are identified discussed and approved to be included.

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Previous List of Principles, will be curated in next version:

#FUF-01	Nature of the Functional Framework
Statement	The Functional Framework is a taxonomy of systems* capabilities that are relevant to the ecosystem of communications service providers. Systems capabilities are what systems are able to provide, defined independent of the context of specific processes, organizations, applications and implementations. *Software and/or hardware
Rationale	It is the intent that all functions of the Service Provider can be supported by (i.e. are able to fit within) the Functional Framework functions. Each function has a detailed description that reflects its purpose.
Implications	The Functional Framework is a catalog of functions/functionalities that are relevant for a Service Provider. The Functional Framework specifies functions along with their descriptions and are grouped in alignment with the Business Process Framework process structure. The functions Process Elements are activity-based, and the Business Process Framework is thus an activity-based process decomposition model.

#FUF-02	Definition of Functional Framework Functions [TO BE EDITED BY KAJ / DONE]
Statement	A function is defined to produce one specified result. Functions are defined at such granularity level that the principle of separation of concerns is considered and when activated the function's result will be accepted without exceptions. Each Functional Framework functions should be unique for each described functionality.
Rationale	The functions of the Functional Framework shall be trusted to be unique. The function shall specify a functionality without exceptions.
Implications	A function specifies the action it will perform. A list of functions can be used to specify a collective functionality of e.g. a component or application, without the risk of conflicting functions or overlapping functionality.

#FUF-03	Information Ownership Functions
Statement	In Domains that are business context specific, (Market_Sales, Customer, Product, Service, Resource and Business Partner) the functions belonging to these domains have a direct responsibility type of relationship (Create, Update and Delete) with Information Framework elements.

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#FUF-03	Information Ownership Functions
	Functions in the Shared Domain can be used in different contexts, which means that mapping to Information Framework elements will differ depending on where the functions are used.
Rationale	A fundamental role of functions is to manage information. Based on the function's purpose it is related to specific information elements.
Implications	Functions are the primary way to manage information and acts on behalf of processes and other functions to manage the information of their ownership. Functions require information from other functions, and they in turn provide information to other functions, but each function should have unique impact on its related information.

#FUF-04	Function Ownership
Statement	A function shall always have a “Function Owner” that acts as the entity or instance that, from an implementation point of view, is responsible for the direct outcome that results from the execution of the function.
Rationale	Responsibility can be shared amongst entities or instances of parties involved in the specification and execution of a particular function, however, accountability–which designates utmost responsibility and ownership of the parent process–refers to the unique instance or person who is ultimately liable for the outcome of the process, the process in turn delegates responsibility to the function in charge for the execution of the task(s) involved. Depending on the implementation scenario, the function can be owned by the delegating process itself, or by a different entity or party (e.g., outsourcing scenario).
Implications	"Shared responsibility is no accountability.” A function is always overseen and “groomed” by its owner who is the person or entity ultimately responsible for the outcome of the execution of the function. If a function owner or accountable party has not been formally specified or designated, the result could be a lack of commitment or accountability in the event that the function is not delivering the expected outcome; likewise, if the function is delivering the expected result, it is the function owner, or the accountable process owner, or both who get credit for the successful outcome.

ADD TEXT ON QUALITATIVE ASPECTS

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#FUF-05	Continuously Optimize Functions
Statement	New functions will be designed and implemented continuously as information systems and technology develop. Functional Framework functions will continuously be analyzed, simplified, or otherwise redesigned as appropriate to follow the development.
Rationale	For the management of the functionality of a corporate's system infrastructure the functions must be continuously improved, and the number of functions expanded. Functions will be used or referred to in many systems architectures. To have access to up-to-date function specifications is of great value. For systems architectures with high flexibility to improve individual functions (e.g., ODA Components) the functional framework and the functions are expected to be updated in a dialog with the industry. For more stable systems the Functional Framework function and the functions are expected to be a source of reference and inspiration.
Implications	There is a need to have an agreed governance of continuous function improvement activity. New technologies will influence the Functional Framework and its functions. Functions must be optimized to align with business development, architecture development and technology development. Additional time and resources will have to be invested in analysis early in the systems lifecycle. Organizational change will be required to implement reengineered systems with optimized functions.

2.2.5. Metrics

(FUTURE WORK)

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3. The Frameworks

3.1. Business Process Framework

The Business Process Framework (also known as eTOM) is an operating model framework for Communication & Digital Service Providers (CSPs & DSPs) in the communications and digital services industries. The model describes the required business processes of service providers and defines key elements and how they should interact.

The Business Process Framework is a standard maintained by the TM Forum, an association for service providers and their ecosystem partners in the communications and digital services industries. It is a hierarchical classification scheme with descriptions of the key business processes required to run a service-focused business. It is also a comprehensive, industry-agreed, multi-layered view of the key business processes required to run an efficient and agile digital enterprise. It provides a common language for use across departments, systems, partners and suppliers, reducing cost and risk of system implementation, integration and procurement.

The eTOM's business-oriented view of the enterprise is useful for planners, managers, strategists and others who need to view the enterprise in business terms, without immediate concern for the nature of the way that these business needs are organized or automated within the business. Therefore, eTOM emphasizes issues such as process structure, process components, process interactivity and the business roles and responsibilities to which these relate.

The Business Process Framework (eTOM) is a reference framework for categorizing all the business activities that a service provider will use in a structured manner that allows these to be addressed at various levels of detail. The processes are grouped by domains and vertical category context and are decomposed starting from high level core, supporting and enabling processes that depict key activities succeeded by lower level unique task activities.

eTOM serves as the blueprint for process direction and provides a reference point for internal process reengineering needs, partnerships, alliances, and general working agreements with other enterprises.

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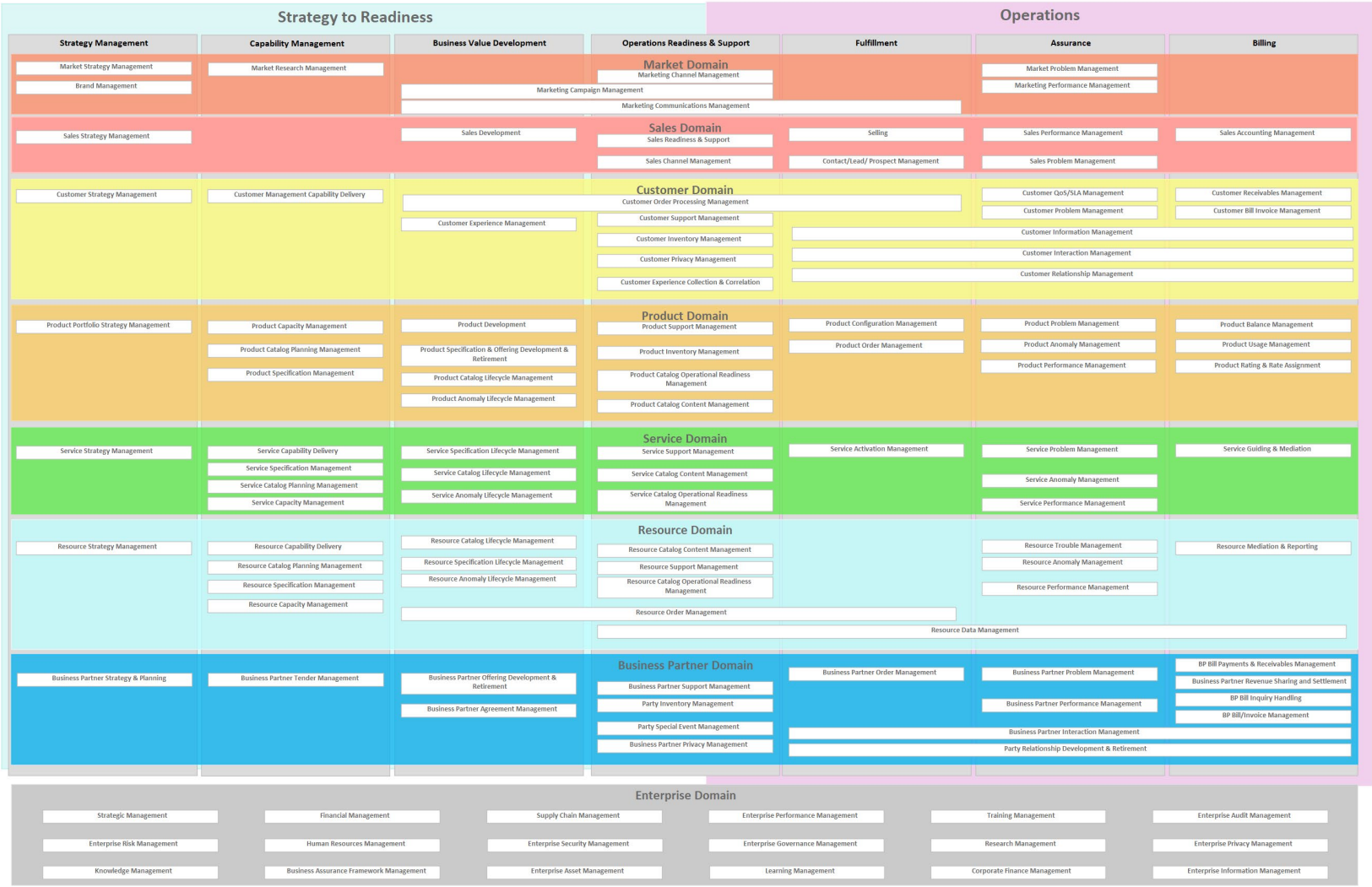

Figure 3-1: The Level 2 map of the currently published Business Process Framework (Please refer to tmforum.org/moda for an up-to-date view)

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3.2. Information Framework

The Information Framework (also known as the SID) provides an information/data reference model and a common information/data vocabulary from a business entity perspective. This 'business view' model uses the concepts of domains and aggregate business entities to categorize business entities, reducing duplication and overlap. Based on data affinity concepts, the categorization scheme is necessarily layered, with each layer identifying in more detail the “things” associated with the immediate parent layer.

The Information Framework's business-oriented view of the enterprise is useful for software engineers, data modelers, enterprise architects as well as business architects and strategists due to the fact that an information model identifies the key data entities in business terms, serving as a bridge between the business and technical areas of a business.

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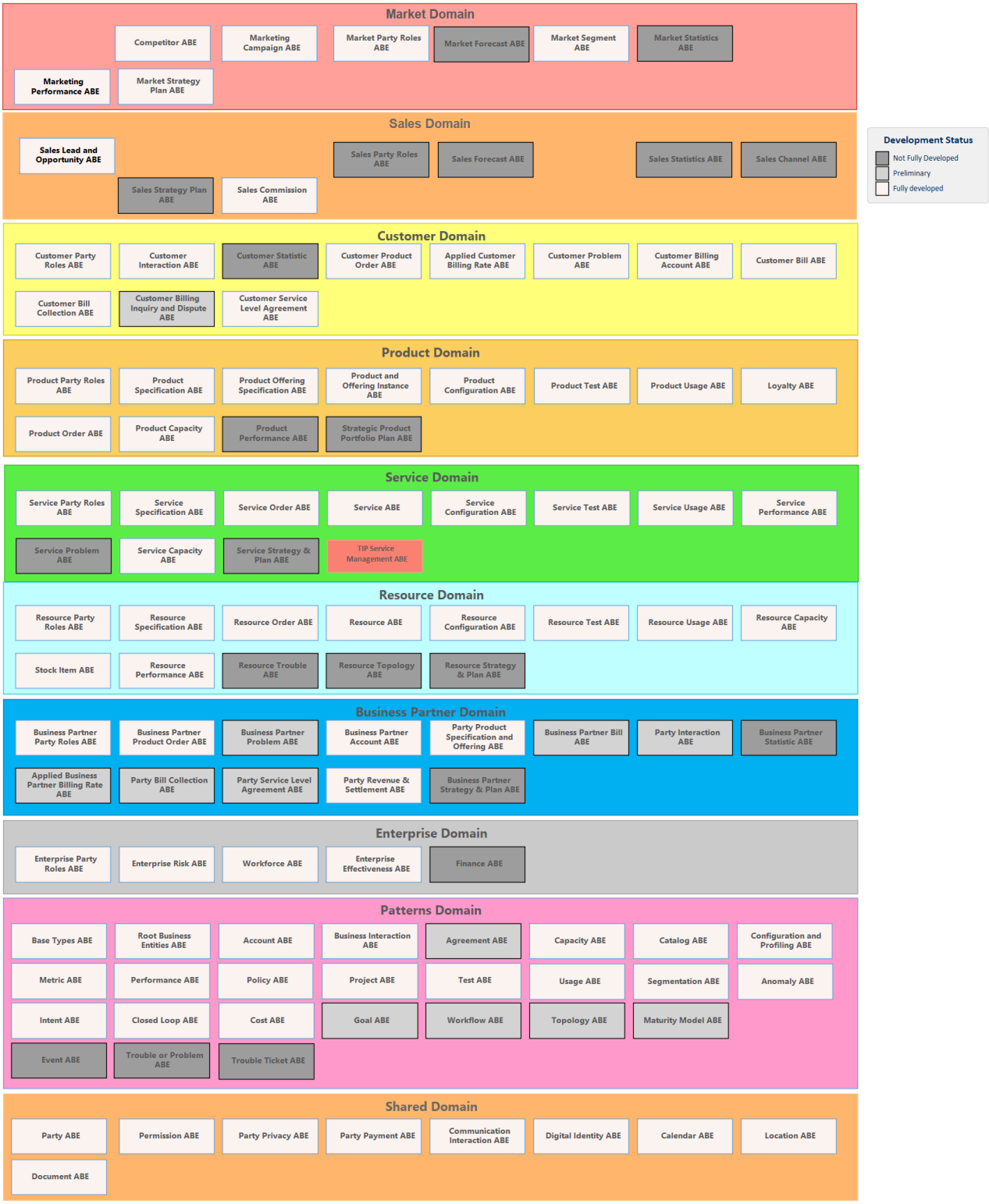

Figure 3-2: The currently published High Level Information Framework (Please refer to tmforum.org/moda for an up-to-date view)

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3.3. Functional Framework

The Functional Framework is a collection of Functions described as automated (computerized) activities, provided for human/machine usage through User Interfaces or machine usage through Integration Services.

The Functional Framework Function is defined to produce one specified result. The ambition is that Functional Framework Functions are defined at such granularity level that when activated the resulting outcome will be completely accepted. For Functions with a description listing more than one sub-function all sub-functions are expected to be performed at every activation.

The granularity level shall also allow only one Function in the Functional Framework for each described functionality. See the figure below for the current high level map of the Functional Framework Functions.

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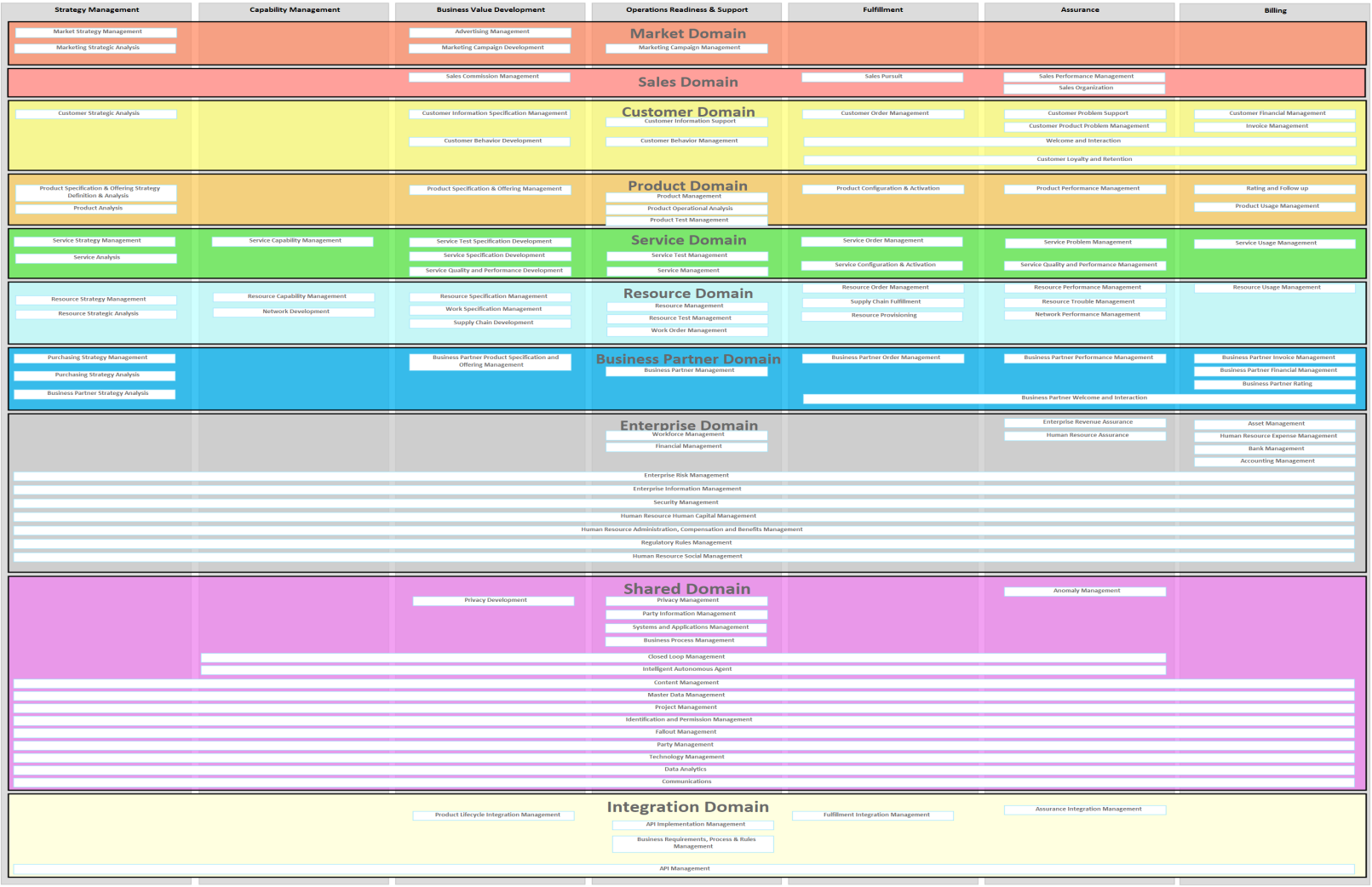

Figure 3-3: The Level 1 map of the currently published Functional Framework (Please refer to tmforum.org/moda for an up-to-date view)

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4. Methodology

This section outlines methodologies for using the frameworks. While the framework provides a singular and definitive structure with rules, concepts and principles to plan or decide assets for the enterprise's information systems, methodologies for modelling target systems and solutions vary and thus this section will highlight relevant methodologies that the frameworks support.

4.1. Business Process Modelling methodology

Business Process Modelling (BPM) is a technique to represent, analyze and improve the activities and workflows of an organization. It helps to identify the goals, inputs, outputs, roles, resources, rules and interactions of each process (including activity and tasks).

By creating a visual and formal representation of the business processes, modelling enables a better understanding of the current state and the potential for improvement. Modelling also facilitates communication and collaboration amongst the stakeholders, such as managers, employees, customers, suppliers and partners (in an ecosystem management play).

Benefits BPM is that it can support the automation, adaptation and governance of the business processes. Automation will use software and/or hardware to perform tasks that would otherwise require human intervention. Automation can increase efficiency, accuracy and consistency of the processes, as well as reduce costs and errors. Adaptation refers to the ability to change or modify the business process models in response to changing needs, demands or opportunities. Adaptation can enhance flexibility, agility and innovation of the organization, as well as enable it respond to/or lead markets. Governance refers to the establishment and enforcement of standards, policies and rules that guide the execution and evaluation of the processes. Governance can ensure quality, compliance and accountability of the organization.

To achieve these benefits, business process modelling leverage standardized catalog of processes, activities and tasks very well scoped, named, defined/described and attributed to their goals. To realize these benefits in practice, BPMs require the use of standards that define the syntax, semantics and notation of the business process models. Standards are important because they enable interoperability, compatibility and integration amongst different entities, tools, systems and platforms that support the modelling, execution and monitoring of business processes. Standards also require and ensure consistency, clarity and completeness of the models, as well as facilitate their validation, verification and testing. Some examples of standards for BPMs are Business Process Model and Notation (BPMN), Business Process Execution Language (BPEL) and Unified Modeling Language (UML).

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- **Business Process Modeling Notation (BPMN):** *This is the most widely used and accepted standard for BPM. It uses a set of graphical elements to depict events, activities, gateways, flows, pools and lanes in a process diagram. BPMN can capture complex and dynamic processes with multiple participants, roles and scenarios. BPMN also supports the execution of process models using Business Process Management Systems (BPMS).*

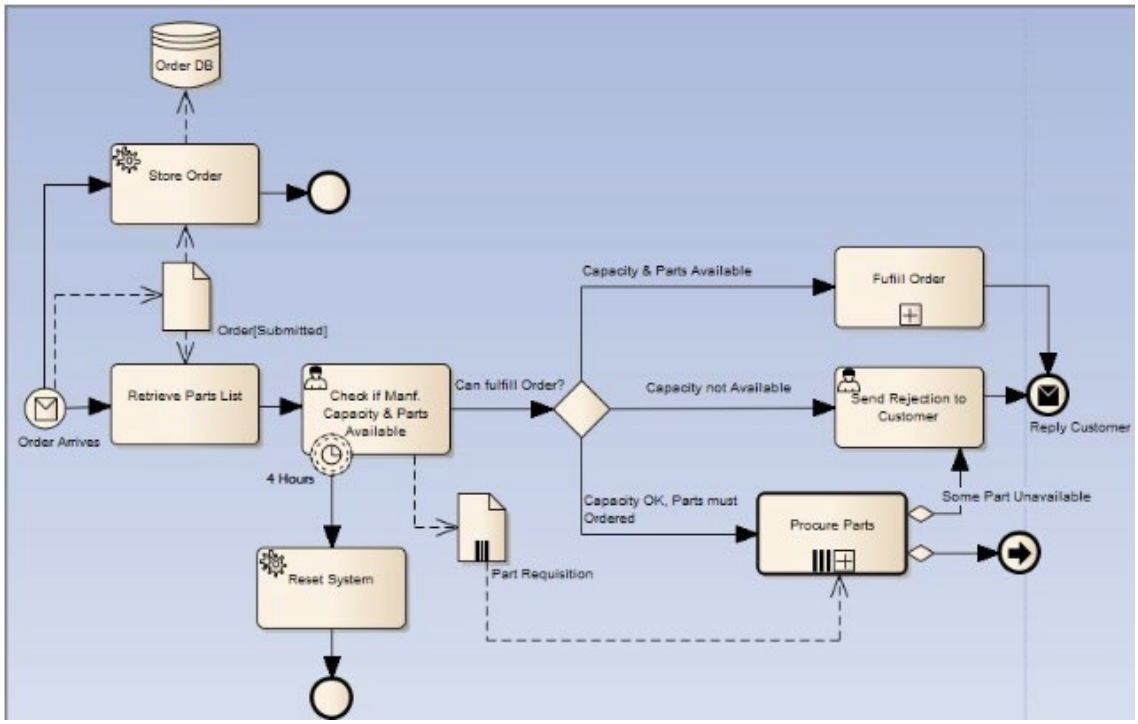

Figure 3-1 Example of BPMN

(reference source: BPMN 2.0 Business Process Toolbox Pages | Enterprise Architect User Guide (sparxsystems.com))

- **Unified Modeling Language (UML):** *This is a general-purpose modeling language that can be used for various domains, including software engineering, systems engineering and business analysis. UML offers 14 types of diagrams to model different aspects of a system or process, such as use case diagrams, activity diagrams, sequence diagrams and state machine diagrams. UML is more suitable for modeling the detailed logic and behavior of a process, rather than the overall flow and structure. UML also requires more technical knowledge and skills to use effectively.*

- **Flowchart:** *This is a simple and intuitive way to model a process using basic shapes and arrows to show the steps and decisions in a workflow. Flowcharts are easy to create and understand, but they have limitations in expressing complex and parallel processes, as well as roles and responsibilities of different actors. Flowcharts are also not standardized, which means they can vary in style and meaning across different contexts and tools.*

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Other models are:

- **Event-driven Process Chain (EPC):** *This is a BPM methodology that uses events, functions, connectors, roles and information objects to model a process. EPC focuses on the triggers and outcomes of each process step, rather than the activities themselves. EPC is often used in conjunction with SAP software to design and implement enterprise resource planning (ERP) systems.*
- **Lifecycle Modeling Language (LML):** *This is a BPM methodology that uses six core elements to model a process: stakeholders, value propositions, actions, measures, decisions and loops. LML emphasizes the value creation and delivery aspects of a process, as well as the feedback loops that enable continuous improvement. LML is designed to be simple, visual and collaborative.*
- **Subject-oriented Business Process Management (S-BPM):** *This is a BPM methodology that uses subjects, messages, states and transitions to model a process. S-BPM views each process participant as an autonomous subject that communicates with other subjects through messages. S-BPM allows for distributed and dynamic process execution, as well as adaptive and agile process design.*

TM Forum's BPF provides the standards for identifying the composition of business processes, a common library of tasks and activities performed with objectives that dovetail the overarching goal of a business process, as well as an attempt to organize them into a formalized framework of the Open Digital Framework. Examples of using the TM Forum BPF can be found in GB921B.

4.2. Information Modelling methodology

The information framework provides a 'data' view of the enterprise. The SID identifies real-world objects of interest, models them as entities (or classes). The framework then identifies important class characteristics, or "attributes". Definitions are provided for both the class and the attributes. In addition, relationships between these classes are identified and modeled as well.

4.3. Functional Framework Modelling methodology

To be added

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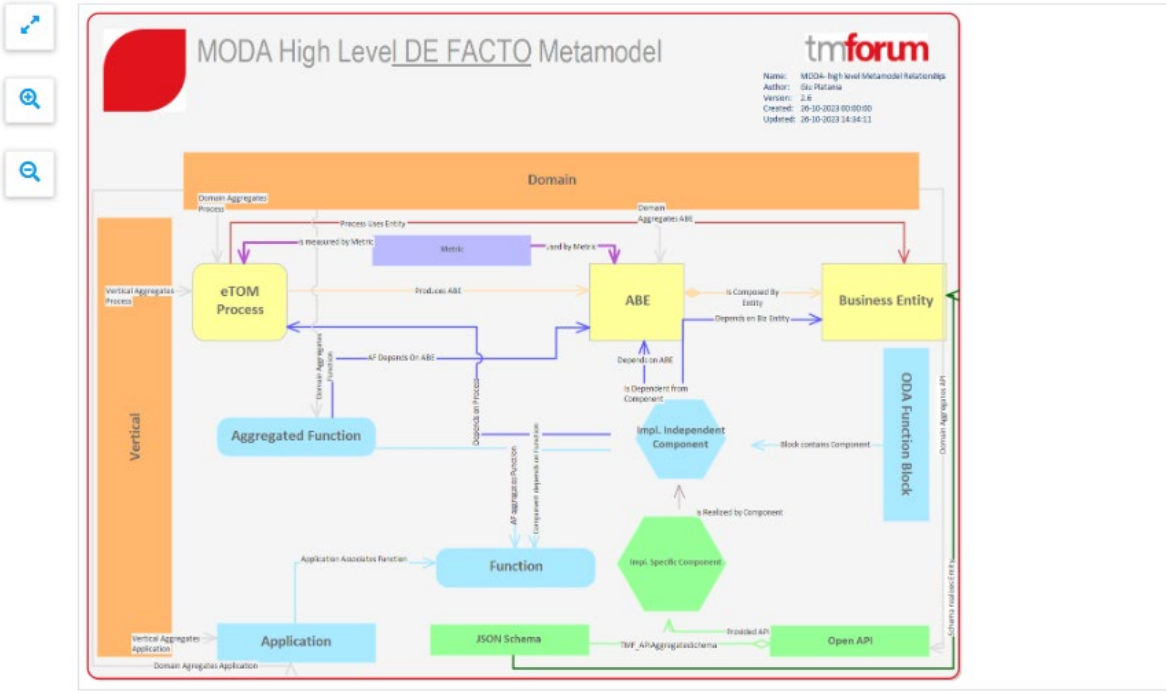
5. Meta Models

This meta-model is the structure of the MODA asset interrelationships as they have been documented in the system-of-record. It has not been formally ratified, hence we refer to it as "de facto".

MODA- high level Metamodel Relationships

TM Forum Metamodel ▸ Metamodel ▸ MODA- high level Metamodel Relationships



Basic Attributes

Name	MODA- high level Metamodel Relationships
Documentation	the TMF MODA high level metamodel is derived from the set of data found in in current forum publications, normalized for consistency. it´s implemented as Model Driven Generation (MDG) technology within the MODA model
Version	2.6
Phase	
Status	
Modified	2023-10-26 14:34:11

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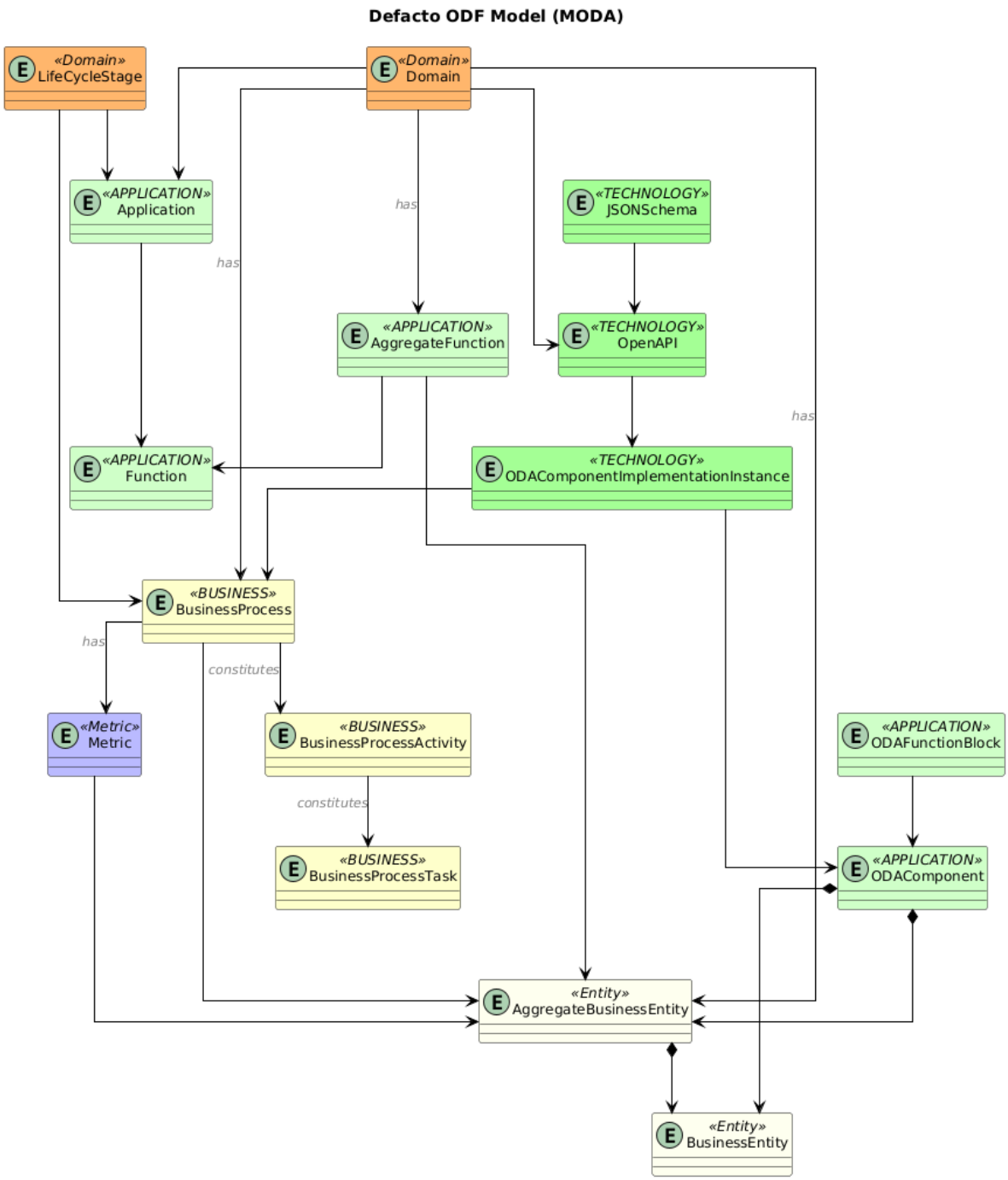

Figure 2 Surrogate model linking key ODF assets.

5.1. Business Process Framework and Information Framework

(FUTURE WORK)

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5.2. Business Process and Functional Architecture

(FUTURE WORK)

5.3. Information Framework and Functional Framework

(FUTURE WORK)

5.4. Business Process Framework and Metrics

(FUTURE WORK)

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6. Administrative Appendix

6.1. About this document

This is a TM Forum Guidebook. The guidebook format is used when:

The document lays out a 'core' part of TM Forum's approach to automating business processes. Such guidebooks would include the Telecom Operations Map and the Technology Integration Map, but not the detailed specifications that are developed in support of the approach.

Information about TM Forum policy, or goals or programs is provided, such as the Strategic Plan or Operating Plan.

Information about the marketplace is provided, as in the report on the size of the OSS market.

6.2. Document History

6.2.1. Version History

Version Number	Date Modified	Modified by	Description of changes
15.5.0	Nov 2015	Frameworx Team (Editor: Alfred J. Anaya)	Initial issue Ongoing merging of Frameworx Concepts & Principles documents from eTOM, SID and TAM
15.5.1	Nov 2015	Alicja Kawecki	Updated cover, minor formatting/cosmetic fixes prior to publishing
15.5.2	Apr 2016	Alfred Anaya	Updated corrected eTOM L2 view with missing processes in the resource domain. Updated slightly the Acronyms section.
16.0 (DRAFT)	Apr 2016	Alfred Anaya	Inserted SID/eTOM mapping section and updated fields and related information accordingly
16.0.1	8 Jun 2016	Alicja Kawecki	Updated cover, header; minor cosmetic edits prior to publication for Fx16
16.5.0	Nov 2016	Avi Talmor	Updated for Rel 16.5
16.5.1	23 Nov 2016	Alicja Kawecki	Minor cosmetic edits prior to publication for Fx16.5
17.5.0	Nov 2017	Avi Talmor	Errata and spelling fixes

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Version Number	Date Modified	Modified by	Description of changes
17.5.1	09 Jan 2018	Adrienne Walcott	Formatting/style edits prior to publishing
18.0.0	June 2018	Andrew Greff	Updated / added definitions:· Application· Application Function· Application InterfaceRemoved definition:· Business Service (as per Frameworx workgroup decision)Added Annex A Definitions / Acronyms / Definitions:· ArchiMate· TOGAF · UMLFormatting:· Defined repeating headers on all tablesUpdated diagrams:· 6-5 eTOM· 10-1 TAMUpdated Section 11:· Merged 11.11 Definitions and 11.12 Application Functions in the Application Framework into Application Function
18.0.1	June 2018	Alfred Anaya-Dubernard	Updated the following· Vertical Process Groupings· Process Areas and Enterprise Management· Common Processes & Patterns· References to Supplier/Partner Domain· Process Descriptions· Process Flow Diagrams (legacy)· Formatting to A4 size
18.0.2	06-Jul-2018	Adrienne Walcott	Formatting/style edits prior to R18 publishing
18.0.1	24-Sep-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status
18.5.0	03-Dec-2018	Cécile Ludwichowski	Update SID map and SID / eTOM mapping
18.5.1	05-Mar-2019	Adrienne Walcott	Updated to reflect TM Forum Approved Status
19.0.0	18-Jun-2019	Cécile Ludwichowski	Update Domains name / descriptions and SID/eTOM mapping
19.5.0	02-Dec-2019	Dirk Rejahl	Updated description of addendum structure. Team Approved
19.5.1	24-Feb-2020	Adrienne Walcott	Updated to reflect TM Forum Approved Status
21.0.0	28-May-2021	Avi Talmor	Updated top level diagram

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Version Number	Date Modified	Modified by	Description of changes
21.0.1	12-Jul-2021	Avi Talmor	Updated to reflect comments received during publishing review cycle
21.0.1	26-Jul-2021	Adrienne Walcott	Updated to reflect TM Forum Approved Status
21.5.0	26-Nov-2021	Avi Talmor Kevin Scaggs	Updated eTOM high level diagram, SID high level diagram, removed Frameworx references, and completed minor eTOM section updates.
22.0	02-Jun-2022	Avi Talmor Cecile Ludwichowski A Anaya-Dubernard Adrienne Walcott Kevin Scaggs	Updated all content and structure to align with changes needed to align with ODA, including Functional Framework chapter and deprecate Frameworx, as well as Application Framework.
22.5	09-Dec-2022	Avi Talmor Kevin Scaggs	Updates to eTOM, Functional Framework, and SID high level diagrams
23.5		Emmanuel A. Otchere	Revamped to clarify usability, updated definitions of key concepts for clarity, updated methodology for applying the framework.
24.6	Dec 2024	Avi Talmor	New diagrams and changes in the FF part
25.5	18-Dec-2025	Avi Talmor	upaded to reflect Market and Sales split

6.2.2. Release History

ReleaseStatus	Date Modified	Modified by	Description of changes
15.5.0	Nov 2015	Frameworx Team (Editor: Alfred Anaya-Dubernard)	Initial release of document Ongoing merging of Frameworx Concepts & Principles documents from eTOM, SID and TAM
16.0.0	Apr 2016	Frameworx Team (Editor: Alfred Anaya-Dubernard)	Inserted SID/eTOM mapping section and updated fields and related information accordingly
16.5.0	Nov 2016	Avi Talmor	Updated for Rel 16.5
17.5.0	Nov 2017	Avi Talmor	Errata and spelling fixes
18.0.0	June 2018	Alfred Anaya-Dubernard	Updates to align with Fx18
18.0.1	24-Sep-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status

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ReleaseStatus	Date Modified	Modified by	Description of changes
18.5.0	03-Dec-2018	Cécile Ludwichowski	Update SID map and SID / eTOM mapping
18.5.1	05-Mar-2019	Adrienne Walcott	Updated to reflect TM Forum Approved Status
19.0	10-Jul-2019	Cécile Ludwichowski A. Anaya-Dubernard	Updated version references, content overhaul for 19.0 including domain names / descriptions, SID/eTOM mapping, diagrams and other graphics and text.
19.5	02-Dec-2019	Dirk Rejahl	Updated description of addendum structure
21.0	8 Jun 2021	Avi Talmor	Updated top level diagram
21.0	26-Jul-2021	Adrienne Walcott	Updated to reflect TM Forum Approved Status
21.5	26-Nov-2021	Avi Talmor A. Anaya-Dubernard Kevin Scaggs	Updated eTOM high level diagram, SID high level diagram, removed Frameworx references, and completed minor eTOM section updates.
21.5	21-Jan-2022	Adrienne Walcott	Updated to reflect TM Forum Approved Status
22.0	02-Jun-2022	Avi Talmor Cecile Ludwichowski A Anaya-Dubernard Adrienne Walcott Kevin Scaggs	Updated all content and structure to align with changes needed to align with ODA, including Functional Framework chapter and deprecate Frameworx, as well as Application Framework
22.5	09-Dec-2022	Avi Talmor Kevin Scaggs	Updates to eTOM, Functional Framework, and SID high level diagrams
22.5	13-Feb-2023	Adrienne Walcott	Updated to reflect TM Forum Approved Status
23.5		Emmauel A. Otchere, A Anaya-Dubernard, Avi Talmor	Overhauled structure, with revamped scope and content to read as a true principles and concept document.
24.6	Dec 2024	Avi Talmor	New diagrams and changes in the FF part
25.5	18-Dec-2025	Avi Talmor	upaded to reflect Market and Sales split

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6.3. Acknowledgments

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